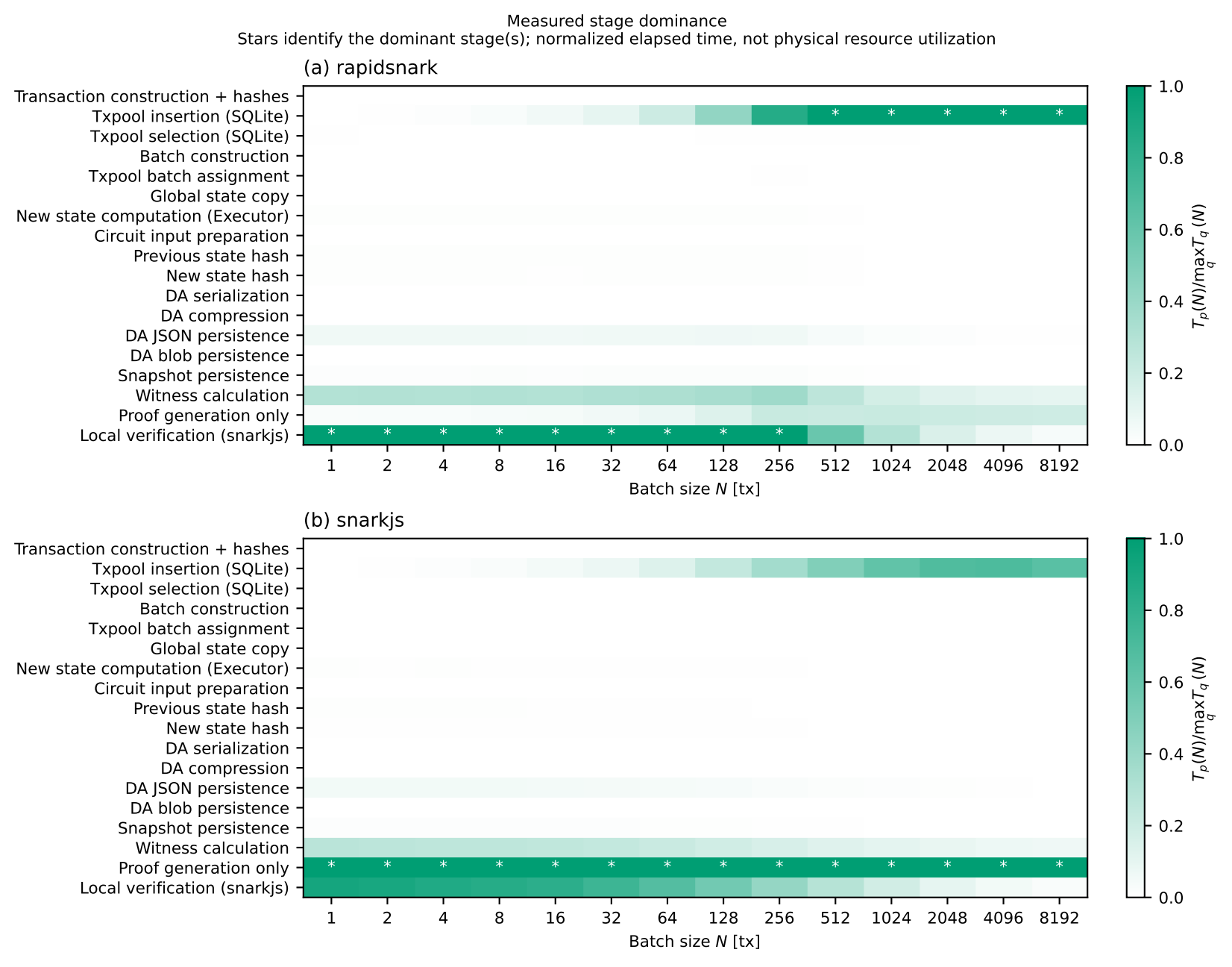
Measured stage dominance
Stars identify the dominant stage(s); normalized elapsed time, not physical resource utilization
(a) rapidsnark
Transaction construction + hashes
Txpool insertion (SQLite)
Txpool selection (SQLite)
Batch construction
Txpool batch assignment
Global state copy
New state computation (Executor)
Circuit input preparation
Previous state hash
New state hash
DA serialization
DA compression
DA JSON persistence
DA blob persistence
Snapshot persistence
Witness calculation
Proof generation only
Local verification (snarkjs)
1 2 4 8 16 32 64 128 256 512 1024 2048 4096 8192
Batch size N [tx]
1.0 0.8 0.6 0.4 0.2 0.0
$T_p(N)/\max_q T_q(N)$
(b) snarkjs
Transaction construction + hashes
Txpool insertion (SQLite)
Txpool selection (SQLite)
Batch construction
Txpool batch assignment
Global state copy
New state computation (Executor)
Circuit input preparation
Previous state hash
New state hash
DA serialization
DA compression
DA JSON persistence
DA blob persistence
Snapshot persistence
Witness calculation
Proof generation only
Local verification (snarkjs)
1 2 4 8 16 32 64 128 256 512 1024 2048 4096 8192
Batch size N [tx]
1.0 0.8 0.6 0.4 0.2 0.0
$T_p(N)/\max_q T_q(N)$